Foundation Models vs Task-Specific Models in Waste Management

A Cross-Domain Evaluation across Classification, Detection, and Segmentation

Guided By: Prof Srijith P.K.

Presented By: Lokendra Mandloi
CS24MTECH11024

Presentation Outline

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- Foundation Models vs Task-Specific Models
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Introduction to Waste Management & Computer Vision

Waste Management Today

- Rapid urbanization has led to a significant increase in solid waste
- Efficient waste handling is essential for:
  - Recycling
  - Resource recovery
  - Environmental sustainability

Role of Computer Vision

- Automates waste understanding from images and videos
- Enables:
  - **Waste classification** – identifying the type of waste
  - **Waste detection** – detecting the presence of waste objects in images/videos
  - **Waste segmentation** – pixel-level separation of waste regions

Foundation Models vs Task-Specific Models

Task-Specific Models

- Trained for a **single task** using **domain-specific labeled data**
- Optimized for **high accuracy** in controlled environments
- Limitation: Poor performance when testing conditions differ from training data

Foundation Models

- Large-scale models trained on **diverse, multi-domain data**
- Can be adapted or used **zero-shot** for multiple tasks
- Advantage: Better **generalization** under domain shift

Key Difference

- Task-Specific → Accuracy-focused under controlled conditions
- Foundational → Generalization-focused

Generalization abilities of foundation models in waste classification

Problem Statement

Problem Statement

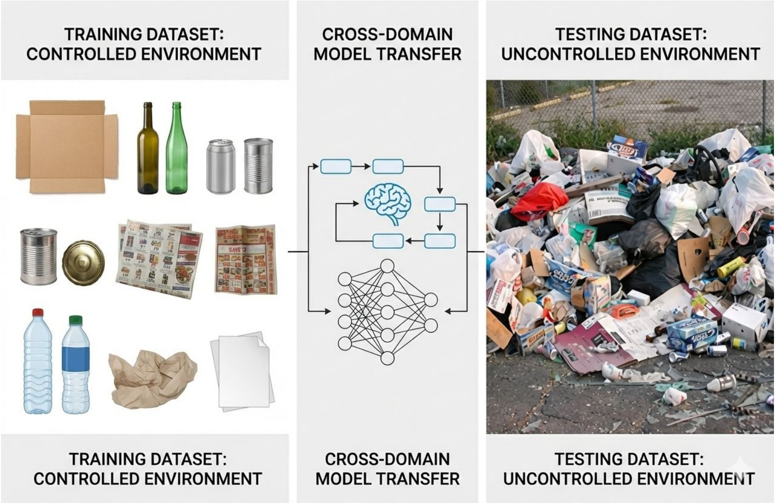
- Most waste detection systems are trained on **clean, curated datasets**
- Real-world waste scenes differ significantly from training data
- This mismatch leads to **unreliable deployment performance**

What is Domain Shift?

- Change in data distribution between **training** and **deployment** data
- Common in waste management due to:
    - Lighting variations
    - Cluttered backgrounds
    - Occlusion and environmental changes

Core Challenge

- High lab accuracy ≠ real-world reliability
- Models must be evaluated under **cross-domain conditions**



Tasks Considered: Classification, Detection & Segmentation

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Waste Classification

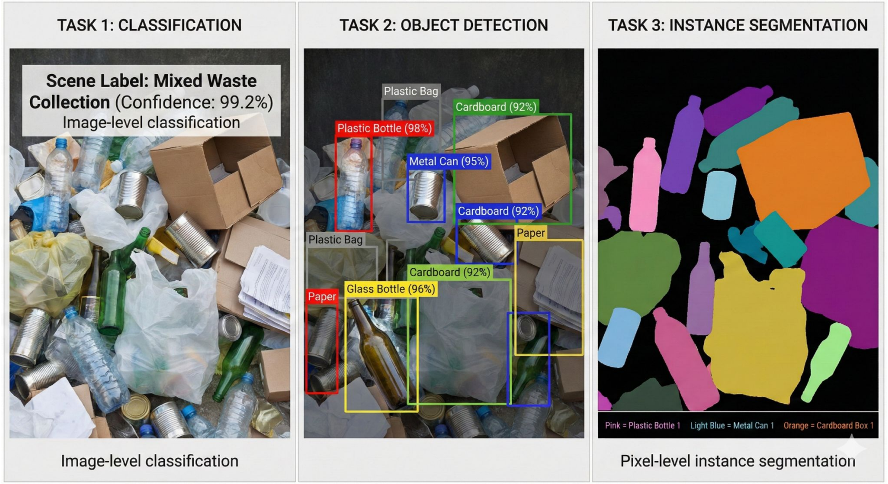
- Identifies the **type of waste**
- Example: plastic, metal, paper, glass

Waste Detection

- Detects the **presence of waste objects** in images or videos
- Determines **which objects are waste**

Waste Segmentation

- Separates waste at the **pixel level**
- Useful for precise localization and sorting



Why Multiple Tasks?

- Real-world waste systems require **different levels of understanding**
- Evaluating all three tasks gives a **comprehensive comparison**

Datasets & Experimental Setup

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Task-Specific (Controlled) Datasets

- **TrashNet** – Lab-based waste classification
- **TACO** – Urban waste detection
- **TACO Masks** – Real-world segmentation masks

Cross-Domain (Real-World) Datasets

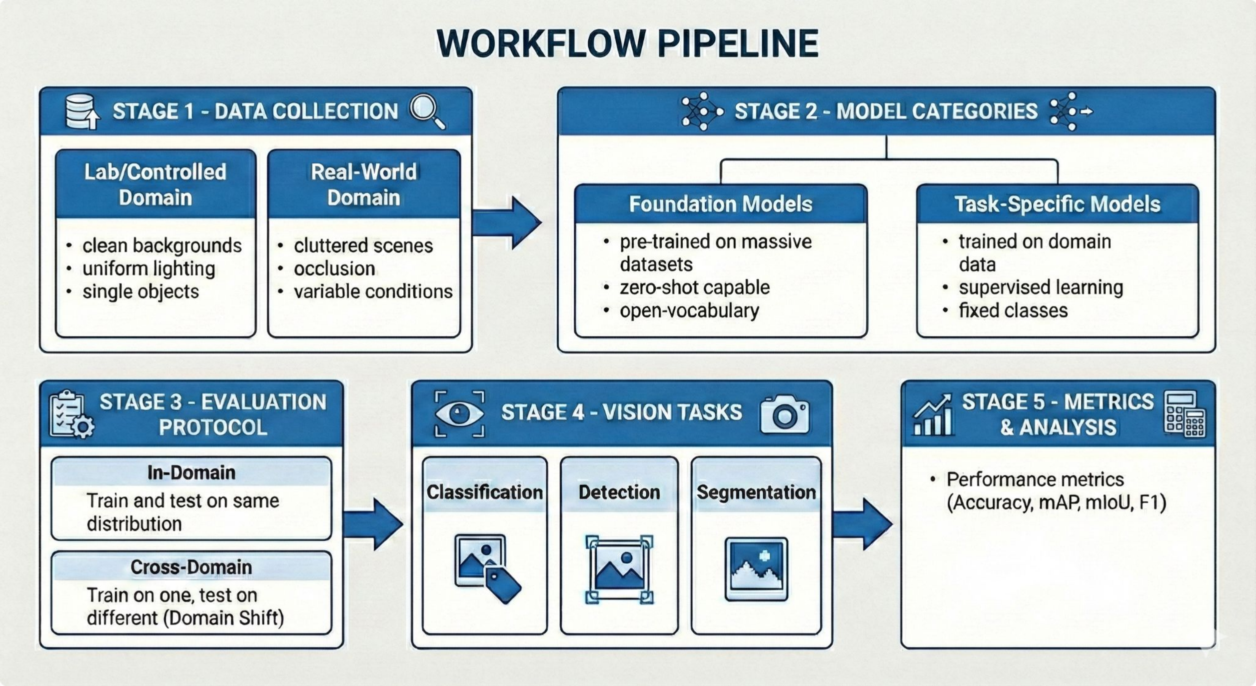
- **RealWaste** – Real-world cluttered waste images
- **Trash-ICRA19** – Underwater waste scenarios
- **Dense Waste Segmentation Dataset** – Controlled segmentation scenes

System Specifications

- **GPU:** NVIDIA A100
- **Training:** 50 epochs

Workflow Pipeline

The workflow is a structured approach to bridging the gap between controlled lab environments and messy real-world conditions, ensuring model resilience through extensive domain-shift analysis.



Experiment Matrix (Models × Tasks × Datasets)

Experiment Matrix : Each model is trained once and evaluated separately on in-domain and cross-domain datasets to quantify domain shift

Task	Model Type	Models	Train Data	Test (In-Domain)	Test (Cross-Domain)
Classification	Task-Specific	ViT-Base, ResNet-50	TrashNet	TrashNet	RealWaste
Classification	Foundation	CLIP	Pretrained	TrashNet	RealWaste
Detection	Task-Specific	YOLOv8, Faster R-CNN	TACO	TACO	Trash-ICRA19
Detection	Foundation	Grounding DINO	Pretrained	TACO	Trash-ICRA19
Segmentation	Task-Specific	U-Net, DeepLabV3+	TACO Masks	TACO Masks	Dense Waste
Segmentation	Foundation	SAM	Pretrained	TACO Masks	Dense Waste

Foundation Model : CLIP

- Jointly trains **image and text encoders** using contrastive learning
- Converts class labels into **text prompts** (e.g., “a photo of a dog”)
- Predicts by matching image embeddings with text embeddings (**zero-shot**)

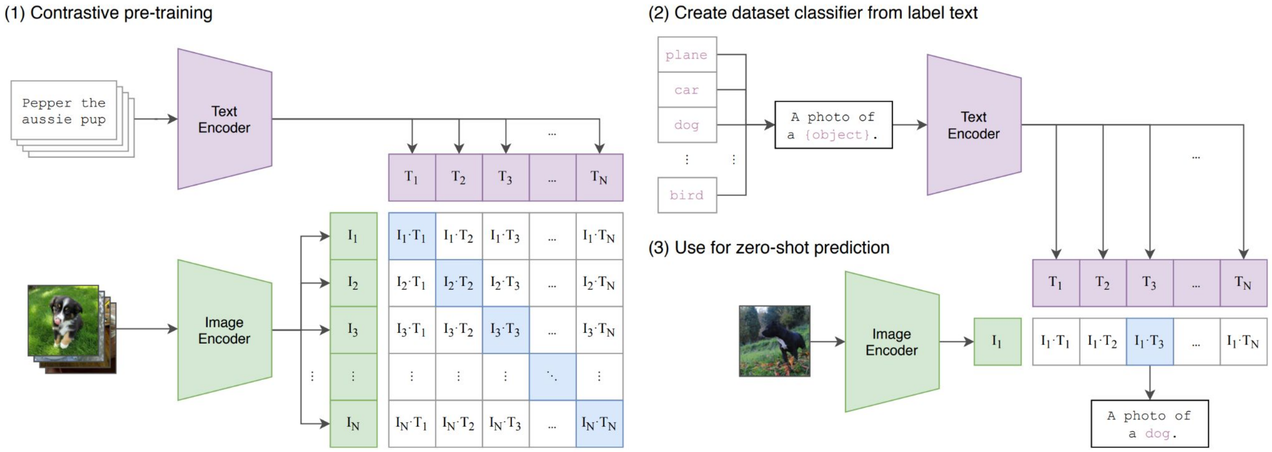


Figure 1. Summary of our approach. While standard image models jointly train an image feature extractor and a linear classifier to predict some label, CLIP jointly trains an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples. At test time the learned text encoder synthesizes a zero-shot linear classifier by embedding the names or descriptions of the target dataset’s classes.

[Learning Transferable Visual Models From Natural Language Supervision](#)

Classification Task

Datasets & Models: TrashNet (task-specific) and RealWaste (cross-domain) evaluated using ViT-Base, ResNet-50 (task-specific) and CLIP (foundation), with Accuracy, Precision, Recall, and F1-score as performance metrics.

Performance Metrics

Dataset	Model	Accuracy	Precision	Recall	F1-Score
TrashNet	ViT-Base	**96.44%**	0.965	0.964	0.964
TrashNet	ResNet-50	80.83%	0.812	0.808	0.806
TrashNet	CLIP	67.83%	0.681	0.678	0.658
RealWaste	ViT-Base	39.98%	0.433	0.446	0.402
RealWaste	ResNet-50	17.85%	0.186	0.239	0.165
RealWaste	CLIP	**42.68%**	0.500	0.539	0.441

Analysis

Task-specific models achieve superior performance in controlled environments, while
CLIP (foundation model) shows better robustness and generalization under real-world domain shift.

Foundation Model : Grounding DINO

- Uses **language-guided queries** to align text and image features
- Applies **cross-modality attention** for text-aware object detection
- Predicts bounding boxes by matching image regions with text descriptions

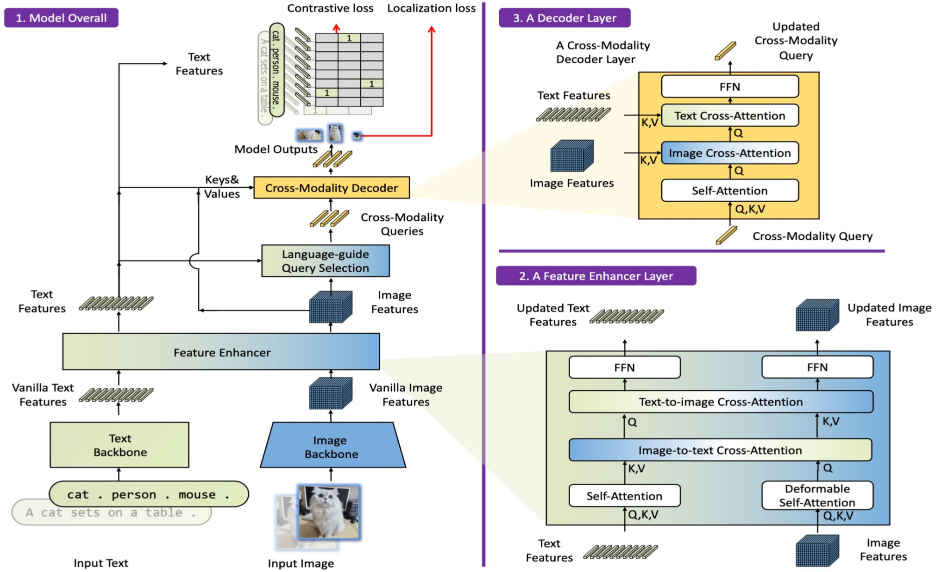


Fig. 3: The framework of Grounding DINO. We present the overall framework, a feature enhancer layer, and a decoder layer in block 1, block 2, and block 3, respectively.

[Grounding DINO: Marrying DINO with Grounded Pre-Training for Open-Set Object Detection]

Object Detection Task

Datasets & Models: TACO (task-specific) and Trash-ICRA19 (cross-domain) evaluated using YOLOv8 and Faster R-CNN (task-specific) and Grounding DINO (foundation), with mAP@0.5, Precision, Recall, and F1-score as performance metrics.

Performance Metrics

Dataset	Model	mAP@0.5	Precision	Recall	F1-Score
TACO	YOLOv8	**64.2%**	0.583	0.712	0.641
TACO	Faster R-CNN	19.3%	0.178	0.215	0.195
TACO	Grounding DINO	15.3%	0.221	0.297	0.253
Trash-ICRA19	YOLOv8	13.5%	0.124	0.158	0.139
Trash-ICRA19	Faster R-CNN	13.6%	0.118	0.162	0.137
Trash-ICRA19	Grounding DINO	**42.7%**	0.256	0.683	0.372

Analysis
Task-specific models perform well in familiar urban environments, while
Grounding DINO demonstrates strong robustness and superior performance under extreme cross-domain conditions.

Foundation Model : SAM

- Encodes the image once using a **powerful image encoder**
- Uses **prompts** (points, boxes, text, masks) to guide segmentation
- Generates **multiple valid masks with confidence scores**

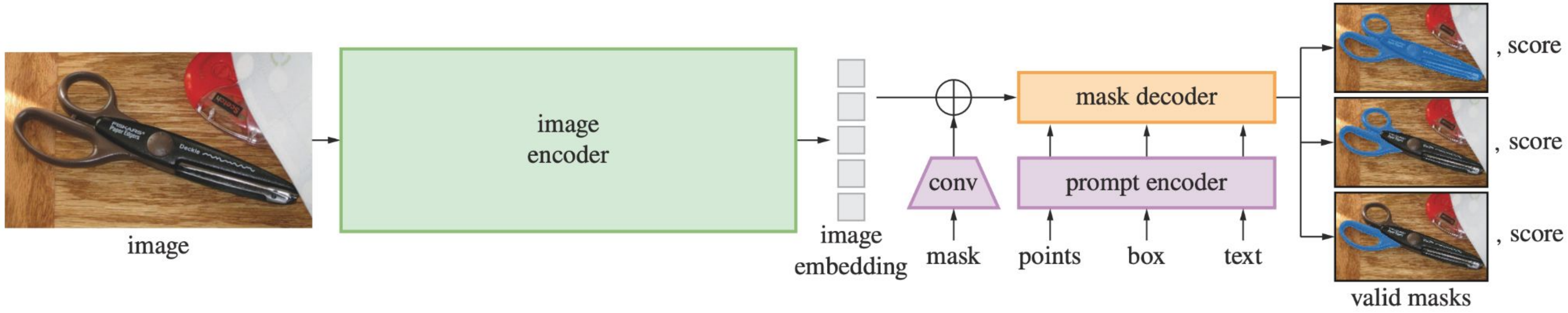


Figure 4: Segment Anything Model (SAM) overview. A heavyweight image encoder outputs an image embedding that can then be efficiently queried by a variety of input prompts to produce object masks at amortized real-time speed. For ambiguous prompts corresponding to more than one object, SAM can output multiple valid masks and associated confidence scores.

Segment Anything

Segmentation Task

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Datasets & Models: TACO Masks (cross-domain) and Dense Waste Segmentation Dataset (task-specific) evaluated using U-Net and DeepLabV3+ (task-specific) and SAM (foundation), with IoU/mIoU, Precision, Recall, and F1-score as performance metrics.

Performance Metrics

Dataset	Model	mIoU	Precision	Recall	F1-Score
TACO Masks	U-Net	32.93%	0.530	0.372	0.398
TACO Masks	DeepLabV3+	**45.41%**	0.818	0.505	0.625
TACO Masks	SAM	3.80%	0.041	0.342	0.073
Dense Waste Segmentation	U-Net	6.30%	0.348	0.065	0.094
Dense Waste Segmentation	DeepLabV3+	4.83%	0.787	0.049	0.092
Dense Waste Segmentation	SAM	**10.23%**	0.130	0.327	0.186

Analysis
Task-specific models excel in controlled segmentation scenarios, while
SAM shows better resilience under real-world domain shift, though segmentation remains challenging overall.

Key Findings

- **Task-specific models** achieve high accuracy in controlled environments but suffer significant performance drops under domain shift.

- **Foundation models** show lower in-domain accuracy but consistently demonstrate **better cross-domain robustness**.

- Domain shift causes **substantial degradation (≈58–80%)** in task-specific models across classification, detection, and segmentation tasks.

Practical Recommendations

- **Controlled environments:** Use **task-specific models** when **high accuracy and low inference time** are required.
- **Real-world deployment:** Prefer **foundation models** for better robustness under **domain shift**, despite higher inference cost.

Future Scope

- **Hybrid Adaptive Cascade:**
  Use fast task-specific models for high-confidence cases and dynamically fall back to foundation models for low-confidence predictions, balancing **accuracy and inference efficiency**.

- **Uncertainty-Aware Domain Adaptation:**
  Integrate uncertainty estimation to identify **when and where models fail** under domain shift, improving reliability in real-world deployment.

- **Multi-Modal Foundation Models:**
  Extend vision-based models by incorporating **additional modalities** (e.g., sensor data or metadata) to improve robustness and decision-making in waste management systems.

Conclusion

- This work presented a **systematic comparison** of foundation and task-specific models across **classification, detection, and segmentation** in waste management.

- Results show that **task-specific models** achieve high accuracy in controlled environments, while **foundation models** demonstrate superior robustness under real-world domain shift.

- The study highlights a clear **accuracy–generalization trade-off**, suggesting that **hybrid and adaptive approaches** are the most practical solution for real-world waste management deployment.

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